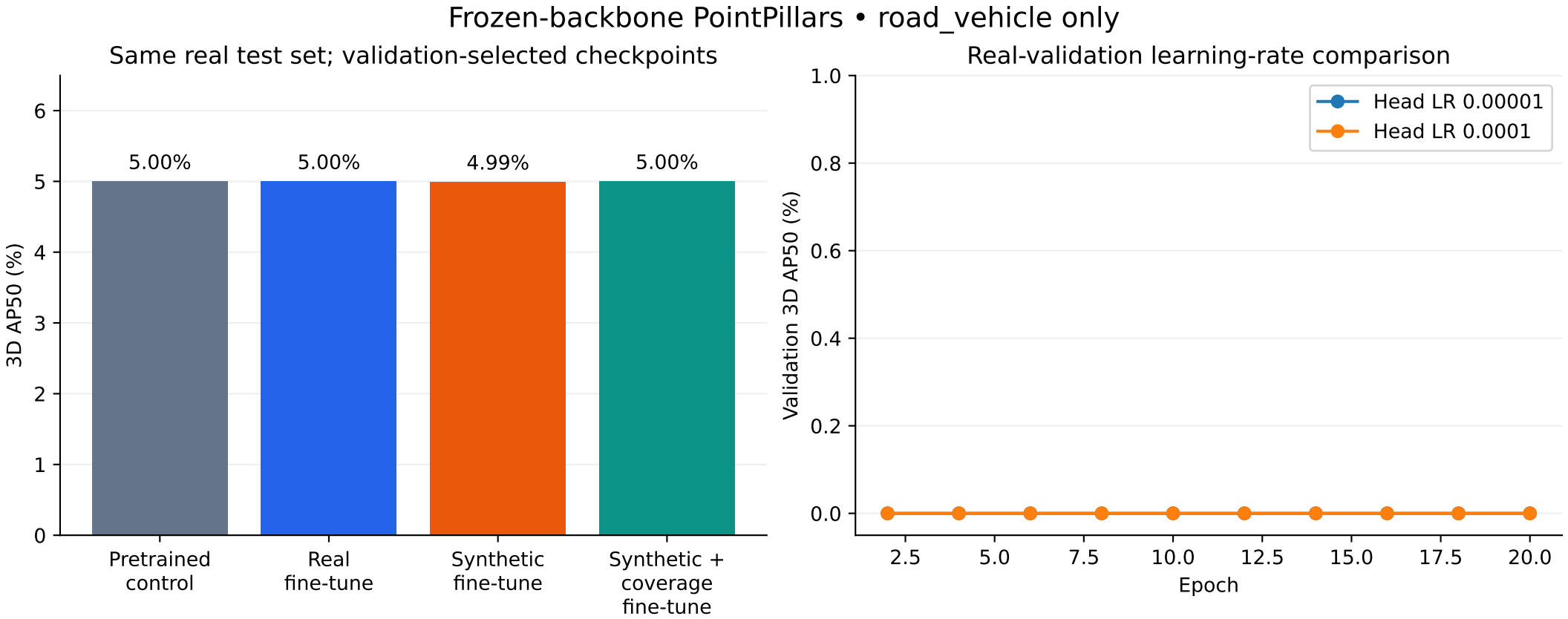
Frozen-backbone PointPillars • road_vehicle only
Same real test set; validation-selected checkpoints
3D AP50 (%)
0
1
2
3
4
5
6
5.00%
5.00%
4.99%
5.00%
Pretrained control
Real fine-tune
Synthetic fine-tune
Synthetic + coverage fine-tune
Real-validation learning-rate comparison
Validation 3D AP50 (%)
0.0
0.2
0.4
0.6
0.8
1.0
Head LR 0.00001
Head LR 0.0001
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Epoch
Known diagnostic fold: 27 real test frames / 86 boxes; one seed. Coverage uses real TRAIN priors.